

Exercicio 1

a)

$$F(A,B,C,D) = \Sigma(1,3,4,6,9,11,12,14)$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	1
11	0	0	1	0
10	0	0	1	1

$$F = A'B'D + AB'D + BC'D' + BCD'$$

$$F = B'D(A'+A) + BD'(C'+C)$$

$$F = B'D + BD'$$

b)

$$F = F(A,B,C) = (A + B') \cdot (C + B + A')$$

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

	00	01
0	1	1
1		1

$$F = A'B' + AC + AB$$

c)

$$Z(A,B,C,D,E,F) = AB + CD + ABDEF$$

	00	01	11	10
00			1	
01			1	
11	1	1	1	1
10			1	

$$Z = AB(1 + DEF) + CD$$

$$Z = AB + CD$$

d)

$$X = ABC$$

	00	01	11	10
0				
1			1	

e)

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$Z = AB + BC' + A'B'C + AB'C'$$

$$Z = B(A+C) + B'(A'C + AC')$$